Other sensors that help detect stages

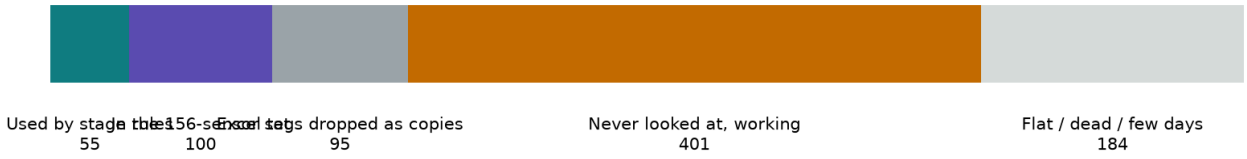
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1. Yes. The operator's commands are the missing sensors.

Today the stages are read from what the plant *does*: stack temperature, current and gas flow. The logger also records what the operator *asked for*: temperature targets, heater setpoints, valve open/close commands and fuel enable switches. None of these are used yet.

These command signals change before the stage shows up on temperature, by 1 to 18 minutes. In the hour before every one of the 11 switches from Hot standby to Thermal adjustment there was a command change. That is the stage pair the rules and the 50-sensor model found hardest.

What was checked: every one of the 835 columns in the 39 daily logger files, not only the 255 Excel tags, averaged per minute and compared with the current stage labels.



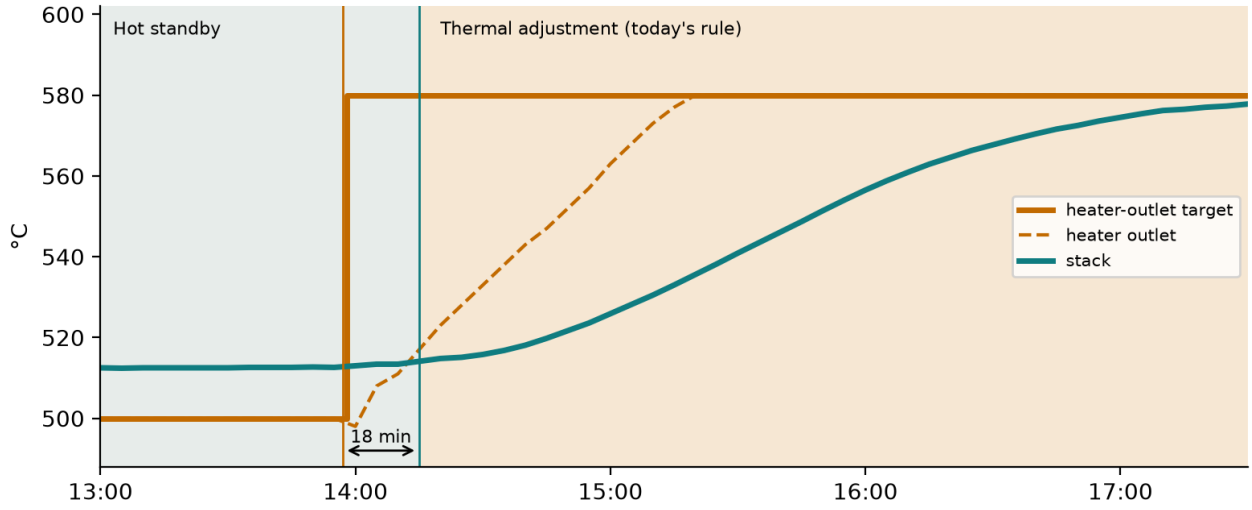
The 401 orange columns had never been looked at. Most are operator commands.

Useful group	Signals	What it shows
Temperature targets	ct_TIRC_206 / 517 SP	The temperature the operator asks the heaters and reformer for. A change comes before every re-heat and warm restart.
Heater power setpoint	EIRC 2,01 setpoint + heater outputs	The heater is turned down 15–18 min before cooling shows on the stack.
Valve commands	MV 1,1,2 · 1,1,3 · 1,2 · 1,3,2 · 1,4,3 · 1,5,3	Air valves open about 6 min before a cold start shows. Fuel valves switch before every load.
Fuel enable bits	FIRC 1,2 / 1,4 enable · PLC fuel values	Natural gas is enabled 9 min before current flows, every time.

If you remember one thing: Temperature tells you a stage has started. The command tells you it is about to start, and why.

2. Commands come first, temperature follows

A stack is heavy ceramic and steel, so it takes minutes to answer a new target. Today's rule waits for that answer. The command is logged the moment it is given.



Real data, 2026-03-29: target raised 500 → 580 °C at 13:57; the rise-rate rule marked Thermal adjustment at 14:15.

For each stage border: did a command change from 30 min before to 10 min after? Same check at 400 random times gives a 0–3 % false-alarm rate.

Border	Seen	Best command signal	Changed at border	How much earlier
Cold start-up (from at ambient)	6	Air valves MV 1,1,2 / 1,1,3	6 of 6	≈ 6 min
Warm restart (from cooling)	3	Heater-outlet target TIRC 2,06	3 of 3	≈ 13 min
Hot standby → Thermal adjustment	11	Any target, heater or valve command	11 of 11 in the hour before	10–15 min (median)
Hot standby → Part load	10	NG enable FIRC 1,2 + fuel valves	10 of 10	≈ 9–12 min
Hot standby → Shut-down	3	Heater controller outputs drop	3 of 3	≈ 18 min
Part load → anything	10	NG enable, fuel valves, PLC fuel values	10 of 10	marks it (short blocks, lead unclear)
Cooling → At ambient	3	Air valves close	2 of 3	≈ 2 min

Part load blocks are 8–44 min long, so lead times there reach back to the load start: these signals mark load, but give no early trip warning.

If you remember one thing: For every start, re-heat and load in the data, a logged command came first.

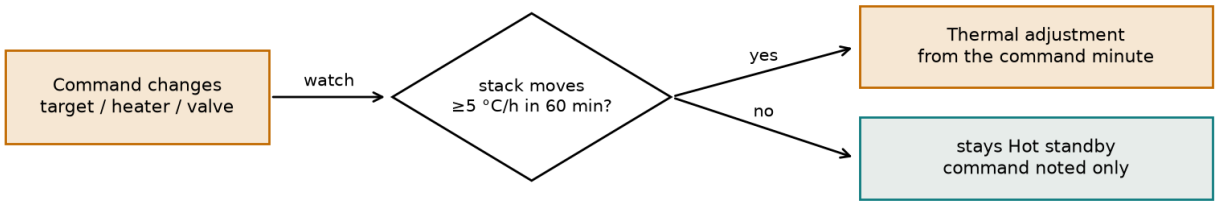
3. Hot standby vs Thermal adjustment

Both stages are hot, unloaded and on forming gas; only the temperature trend differs. No temperature, pressure or monitor column separates them (best 0.3 on a 0–1 scale); the 50-sensor model got about half of Thermal adjustment right. The commands show why the temperature starts to move.

Start (2025/26)	Operation	Command in the hour before
01-31 18:33	Cool-back	heater ↓
02-02 13:57	Dip during hold	heater ↓
02-18 10:22	Re-heat	target 500→600
03-03 14:33	Re-heat	target 500→570
03-04 11:27	Fuel switch-over	fuel valves
03-04 14:42	Re-heat	reformer 800→700
03-20 15:01	Cool-back	heater setpoint 21→54
03-24 08:06	Re-heat	target 570→585
03-24 13:22	Cool-back	target → 0
03-29 14:15	Re-heat	target 500→580
03-30 08:38	Fuel switch-over	fuel valves

Command	Given in Hot standby	Stage changed within 60 min
Heater-outlet target TIRC 2,06	14	13
Heater power setpoint EIRC 2,01	16	12
Fuel / forming-gas valves	13	12
Reformer target TIRC 5,17	6	4

How it would be used: the command sets the start time, the temperature confirms it.



If you remember one thing: All 11 Thermal adjustments had a logged reason: a raised or lowered target, a heater change or a fuel switch.

4. What did not help, and the next step

Group	What the data shows	Advice
Possible room / ambient sensors QIR 10,23 · 10,24 · 10,25	Warm up with the plant (10,25 reads 6–11 K above the probe proxy when the stack is hot) and drop to 0 °C in data gaps. 10,24 is closest (r 0.64 to the proxy). They most likely sit inside the enclosure, not outdoors.	Keep the probe proxy. Ask the site where these three are mounted.
Insulation monitors B0–B4 (170 columns)	Logged on 19 of 39 days only. They separate load from no load a little and do not help with any hard pair.	Leave out.
DC converter mode / off status EIRC 4,x	Same information as string current. Switches to 'run' 0–7 min before current flows (12 loads).	Optional: use for the 'Pre-load' minutes.
95 temperature / pressure tags dropped as copies	They separate hot from cold just like stack temperature. They add nothing for Hot standby vs Thermal adjustment.	No change.
Fieldbus, safety-PLC and UPS states	Only 3–4 days logged.	Leave out.
Idle (warm plant off)	Only one day (2025-10-30) in the data, so no sensor can be tested for it.	Needs more data.

Proposed next step

1. Add command signals. Load the 4 groups in stage_signals.py: TIRC 2,06 / 5,17 targets, EIRC 2,01 setpoint, MV valve commands, FIRC 1,2 enable.

2. Command marks the start. In operating_states.py and ladder_stages.py: start Start-up, Thermal adjustment and Shut-down at the command minute when temperature confirms within 60 min.

3. Check the effect. Re-run flow_check and the 50-sensor test. Report how many minutes move and whether Thermal adjustment detection improves.

Your decisions

1. Borders: move stage starts to the command minute, or only add a 'commanded at' column?
2. Gas phase: read it from valve commands instead of flow meters?
3. Ambient: where are QIR 10,23 / 10,24 / 10,25 mounted?

If you remember one thing: The useful new signals are commands. The extra measurements mostly repeat what stack temperature already says.

5. Which other flows keep varying?

All 25 flow-type columns were checked. Outside the flows the rules already read, only three vary in a way that matters: the cathode air level, the anode ejector pressure difference and the 3-way air valve. The air level is the useful new one: it shows whether a cool-down is forced (8 000–10 000 ln/min, about 3× faster) or normal (6 000).

Signal	Status	Cold off	Start-up	Hot standby	Thermal adj.	Part load	Cooling
Cathode air FIRC 1,1,1 + 1,1,2	rules use on/off only	0	6000	6000 · 10000	6000 · 10000	9000	1000 → 10000
Forming gas N_2 1,3 + H_2 1,5	in rules	0	1200 + 60	1200 + 60	1200 + 60	0	1200 + 60
FIRC 1,2 = POx air	rules call it NG	0	0	0	0	320	0
FIRC 1,4 = natural gas	not in rules	0	0	0	0	120	0
Anode ejector Δp	not in rules	−0.2	0 · 2.6	3.4	3.5	16	0 · 2.3
3-way valve output (FIRC 1,1,3)	only from 03-17	0	0–30 %	0	0–40 %	0–39 %	0–50 %
PLC fuel values (AFSR, POx, FU, ejector p)	not in rules	0	0	0	0	on	0

Shut-down cooling, stack above 150 °C, by total cathode air:

Air, ln/min	Typical cooling rate	When
1 000	−4 °C/h	slow tail at the end of cooling
6 000	−14 °C/h	normal level, most of the time
8 000	−41 °C/h	forced cooling (late March)
9 000	−51 °C/h	forced cooling
10 000	−41 °C/h	forced cooling

Ejector Δp confirms anode gas is circulating (≈3.4 forming gas, ≈16 under load) and repeats the gas phase. The 3-way valve output opens under load (35 %) and in cooling (10 %), logged only from 2026-03-17.

Two labels are wrong in the current code

Issue	Evidence
FIRC 1,2 is POx air, not natural gas	Reads ≈ 320 ln/min, identical to the PLC's own POx value. The real gas meter FIRC 1,4 reads ≈ 120 (range 0–200). Both switch together, so stage borders do not change, but operating_states.py and the fake-data generator label POx air as NG.
'FIRC 1,1,3' from 2026-03-17 is a temperature	The logged value equals TIR 2,05 (EAPH inlet) to 0.0 °C. FIRC 1,1,3 is the 3-way air valve, controlled on that temperature (target 570/590 °C). The generator treats it as a flow in ln/min. The real valve movement is its output, 0–50 %.
Old FIRC 1,1,3 tag (before 03-17)	Reads ≈ 0.16 at all times: noise, no information.

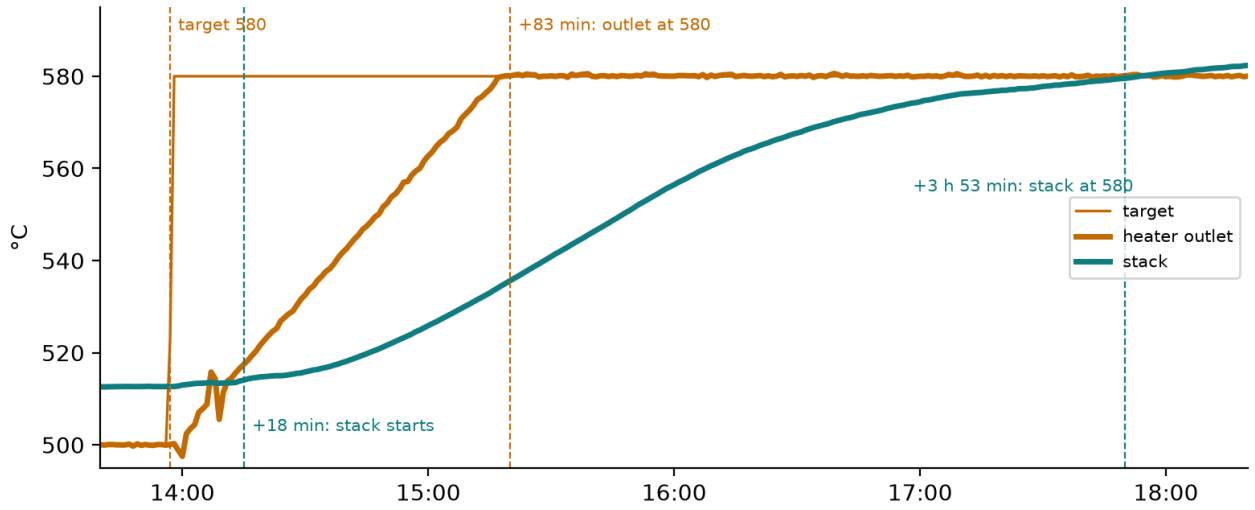
Decisions: 1. Relabel FIRC 1,2 as POx air and read NG from FIRC 1,4? 2. Generator: replace 'FIRC 1,1,3 ln/min' with valve output %, or drop it? 3. Split Shut-down by air level (forced ≥ 8 000 vs normal)?

If you remember one thing: Of the flows, only the air level adds stage information. Two flow tags are mislabelled and should be fixed first.

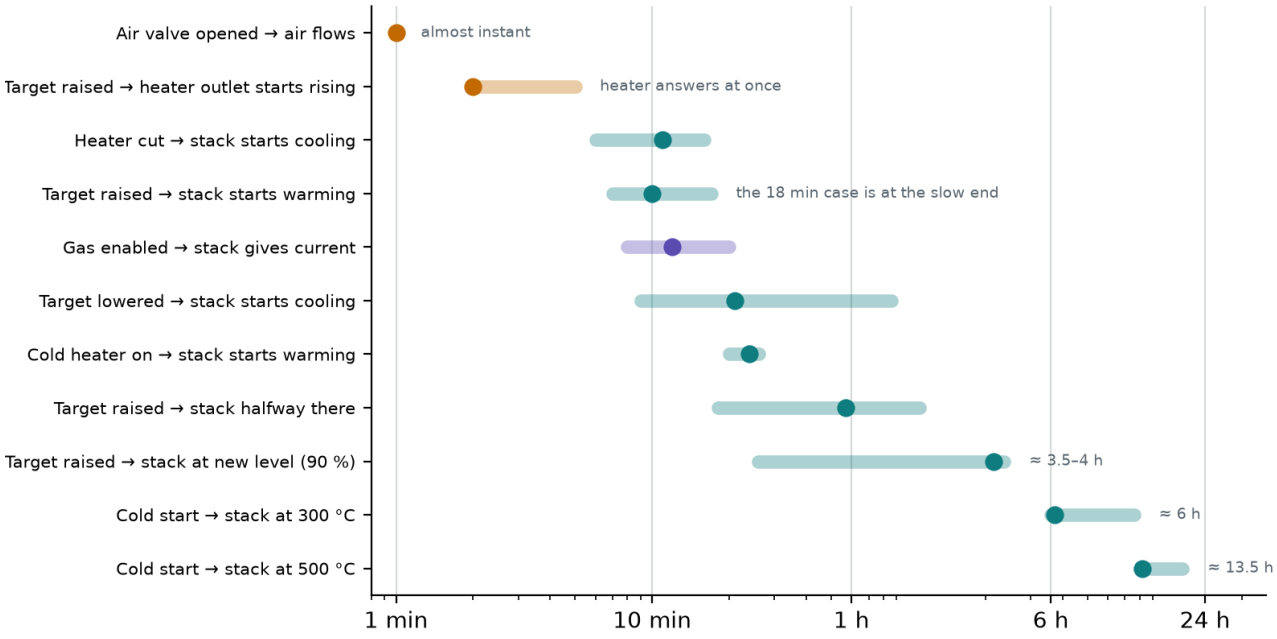
6. Response times: 18 minutes is when the stack starts to warm. Reaching the new temperature takes about 4 hours.

The plant works as a chain: the operator gives a command, the heater or valve reacts, and the heavy stack follows much later. Each delay was measured on every matching event in the 39 days of logs.

In the 18-minute example the stack only started to warm 18 minutes after the target was raised. The heater outlet reached the new target after 83 minutes, the stack after almost 4 hours. From cold, the stack needs about 6 hours to reach 300 °C and about 13.5 hours to reach 500 °C.



Real data, 2026-03-29. The heater outlet climbs at a fixed ramp of about 1 °C per minute, so the stack cannot catch up faster.



Dot = typical delay (median), bar = middle half of the events. Orange = valve or heater, teal = stack temperature, purple = electrical output.

What happens → what follows	Events	Typical	Usual range
Air valve opened → air flows	22	1 min	1 min–1 min
Target raised → heater outlet starts rising	11	2 min	2 min–5 min
Heater cut → stack starts cooling	8	11 min	6 min–16 min

What happens → what follows	Events	Typical	Usual range
Target raised → stack starts warming	10	10 min	7 min–17 min
Gas enabled → stack gives current	10	12 min	8 min–20 min
Target lowered → stack starts cooling	21	21 min	9 min–1.4 h
Cold heater on → stack starts warming	8	24 min	20 min–26 min
Target raised → stack halfway there	9	57 min	18 min–1.9 h
Target raised → stack at new level (90 %)	9	3.6 h	26 min–4.0 h
Cold start → stack at 300 °C	7	6.2 h	6.0 h–12.7 h
Cold start → stack at 500 °C	6	13.6 h	13.3 h–19.6 h

Three patterns

Fast (1–2 min): valves and the heater obey at once. **Medium (10–24 min):** the stack starts to move, which is why temperature rules lag behind commands. **Slow (hours):** a 60–80 °C re-heat takes 3.5–4 hours; cooling 50 °C after the heater is cut takes about 4 hours.

Load heats the stack fast

Once current flows, the stack warms 2 °C within 2–5 minutes and keeps climbing about 1–2 °C per minute, faster than the heater can push it. All 10 load runs in the data lasted 44 minutes or less.

Load run start	Minutes on load	Stack rise, °C	°C per min
2026-03-24 08:54	32	+58	1.8
2026-03-24 13:54	43	+47	1.1
2026-03-25 14:30	44	+90	2.0
2026-03-19 15:31	24	+39	1.6
2026-03-13 13:26	20	+30	1.5

No clean delay: raising cathode air to 10 000 ln/min while hot (heater usually changed at the same moment); reformer target (mostly switched on rather than stepped; its temperature starts moving after about 8 minutes).

Uses and decisions: 1. Build these delays into the fake-data generator? 2. Use 'command, then stack moves within 30 min' as a check in the stage rules? A stack that answers much slower than usual could also flag heater or insulation wear.

If you remember one thing: Valves and heater answer in 1–2 minutes, the stack starts to move after 10–25 minutes, and it arrives hours later.

Web version: https://claude.ai/artifact/HikmTKCvpJXydM5S43hAh7